Golden Gate Assembly

REACTION — 10 µL

Pipette top to bottom. **Enzymes last.**

6 ddH_2O (white rack)
1 10× T4 DNA ligase buffer (red)
1 each DNA fragment (2 of them, 2 µL total)
0.5 T4 DNA ligase
0.5 BsaI

10 µL total

10 µL is deliberate — it is exactly what the transformation takes. Carry the whole reaction forward; do not split it.

THEN

1. Mix well and quick spin.
2. Label the tube with your assigned code.
3. Put every reaction from your section in **one thermocycler block**.
4. Run the program below.

PROGRAM — 2 INSERTS

```
repeat 30×:
  37 °C — 1 min
  16 °C — 1 min
60 °C — 5 min
4 °C — hold
```

NEB's recommended protocol for the Golden Gate Assembly Kit (#E1601).

37 °C is BsaI cutting, 16 °C is T4 ligase sealing. Cycling drives the mixture toward the assembled product, which no longer has a BsaI site to cut.

BY INSERT COUNT (NEB)

- **1 insert** — 37 °C 5 min (or 1 h for library prep) → 60 °C 5 min
- **2–10 inserts** — (37 °C 1 min → 16 °C 1 min) × 30 → 60 °C 5 min
- **11–20+ inserts** — (37 °C 5 min → 16 °C 5 min) × 30 → 60 °C 5 min

NOTES

- **The closing 60 °C is not an inactivation step.** It favours cutting **without** ligation, so destination plasmid that was never cut, or that religated, gets linearised. That is what keeps background colonies down. Do not skip it.
- **The ligase buffer supplies the ATP.** It is the component most often forgotten, and without it nothing ligates.
- Mix the DNAs equimolar if you can. If your preps are consistent, do not bother normalising.
- Miniprepped, gel-purified and Zymo-cleaned DNA all work.
- Scaling up or down is fine as long as the buffer ends at **1×** and the DNA is not so concentrated that it inhibits the ligase.

> **GG1 is the same idea, different numbers.** 25 × (37 °C 2 min → 16 °C 5 min), then 45 °C 10 min and 80 °C 10 min. The cycling is fine. The ending differs: **80 °C kills BsaI**, so vector that was never cut or that religated survives to transform. Expect more background.